

LANGUAGE MODELS CHEATSHEET

AI Engineer Ready · Week 2 — What an LLM Actually Predicts

Golden Rule → text → tokens → predict next token → repeat → decode back to text

1. What Is a Language Model?

A language model estimates the probability of the next token given everything before it: $P(w_t | w_1, w_2, \dots, w_{t-1})$. Modern LLMs (GPT, BERT, Claude, etc.) generate or understand text by working with this single objective, applied one token at a time.

```
$ prompt: "The capital of France is"
$ model predicts a probability distribution over the whole vocabulary:
$ "Paris" -> 0.87
$ "a" -> 0.02
$ "located"-> 0.01 ...
$ highest-probability (or sampled) token is appended, then the process repeats
```

2. Where Language Models Show Up

Use case	What next-token prediction gives you
Autocomplete	Suggests the most likely next word(s) as you type
Chatbots / Q&A	Generates a full reply one token at a time
Summarization	Predicts a shorter token sequence that compresses the input
Code assistants	Predicts the next line/token of code from context
Search / ranking	Scores how likely a query and a document belong together

3. How We Got Here (Evolution)

```
$ n-gram models -> RNN / LSTM -> Transformers -> modern LLMs
$ (count word (remember a (attend to ALL (GPT, BERT, Claude,
$ sequences) short history) tokens at once) Gemini, Llama...)
```

What they learn from: large corpora of books, websites, articles, code, and conversations. **How they work, in one line:** convert text to tokens → predict next-token probabilities → generate (or classify) token-by-token.

Remember

A model doesn't "know" facts the way a database does — it has learned statistical patterns of what token is likely to come next. That's why it can sound confident while being wrong (see the LLM Limitations cheatsheet).

4. Key Terms Quick Reference

Term	Meaning
LLM	Large Language Model — a language model with billions of parameters trained on huge text corpora

Term	Meaning
Autoregressive	Generates output one token at a time, each new token conditioned on all previous ones
Parameters	The learned weights inside the model; "large" usually means billions of them
Inference	Running the trained model to produce an output (as opposed to training it)
Context window	The maximum number of tokens the model can "see" at once (prompt + response)